FLEx Tutorial: From Excel to HTML Dictionary

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This tutorial provides a simple workflow for creating a digital dictionary using FLEx (FieldWorks Language Explorer). It shows how to convert lexical data from Excel into a structured format, import it into FLEx, and export it as a searchable HTML dictionary. The process is suitable for beginners and supports the development of trilingual dictionaries such as Varkani–Persian–English for research and community use.

Step 1: Prepare your Excel file

The first step is to **sort the data in the Excel file**. Since my dictionary is trilingual, I organized the columns accordingly.

	A	B	C	D	E	F
1	**ID**	**Varkan**	**Persian**	**English**	**PartOfspeech**	**Domain**
2	1	ašgamdâr	آبستن	Pregnant	noun	body
3	2	čölm	آب بینی	Nasal discharge	noun	body
4	3	töf	آب دهان	Saliva	noun	body
5	4	künâľjena	ارنج	Elbow	noun	body
6	5	öroq	آرغ	Saliva (mouth discharge)	noun	body
7	6	abrü	ابرو	Eyebrow	noun	body
8	7	ostoxun	استخوان	Bone	noun	body
9	8	qelam-e pâ	استخوان ساق پا	Shin bone	noun	body
10	9	ašg	اشک	Tear	noun	body
11	10	angošt	انگشت	Finger	noun	body
12	11	sabbâba	انگشت سبابه	Index finger	noun	body
13	12	šast	انگشت شست	Thumb	noun	body
14	13	qalič	انگشت کوچک	Little finger	noun	body
15	14	bâzü	بازو	Arm	noun	body
16	15	bädan	بدن	Body	noun	body
17	16	band angöšt	بند انگشت	Finger joint	noun	body
18	17	xâyä	بطن	Abdomen	noun	body
19	18	damâq	بینی	Nose	noun	body
20	19	pâ	پا	Foot	noun	body
21	20	küne pâ	پاشنه پا	Heel	noun	body

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Step 2: Convert Excel (CSV) to SFM

Next, the Excel (or CSV) file needs to be converted into **.SFM format**, which is compatible with FLEx.

To convert the file to **.SFM**, you can use one of the following methods:

- **Method 1: Using Python**

Since Python was already used, this method is efficient and reliable.

Create a file named:

excel_to_sfm.py

Then paste the following code:

```
import csv

with open("dictionary.csv", encoding="utf-8") as f:
    reader = csv.DictReader(f)

    with open("dictionary.sfm", "w", encoding="utf-8") as out:
        for row in reader:
            out.write(f"\lx {row['Headword']}\n")
            out.write(f"\ge {row['English']}\n")
            out.write(f"\gf {row['Persian']}\n\n")
```

Run the script with:

```
python excel_to_sfm.py
```

- **Method 2: Using AI**

You can also use AI tools by prompting:

Convert this CSV into SFM format for FLEx

However, it is important to carefully review the output to ensure the formatting is correct.

In this project, I used the second method.

Step 3: Open FLEx

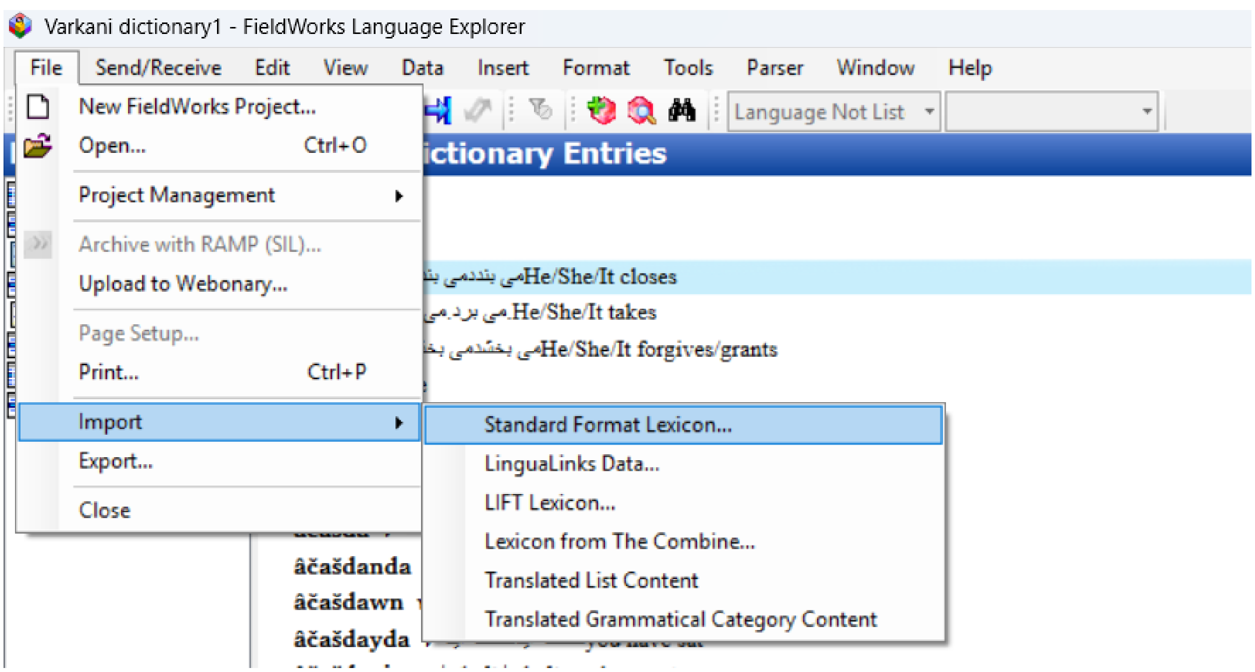
Open FLEx and create a new project or open an existing one.

Step 4: Import the SFM file

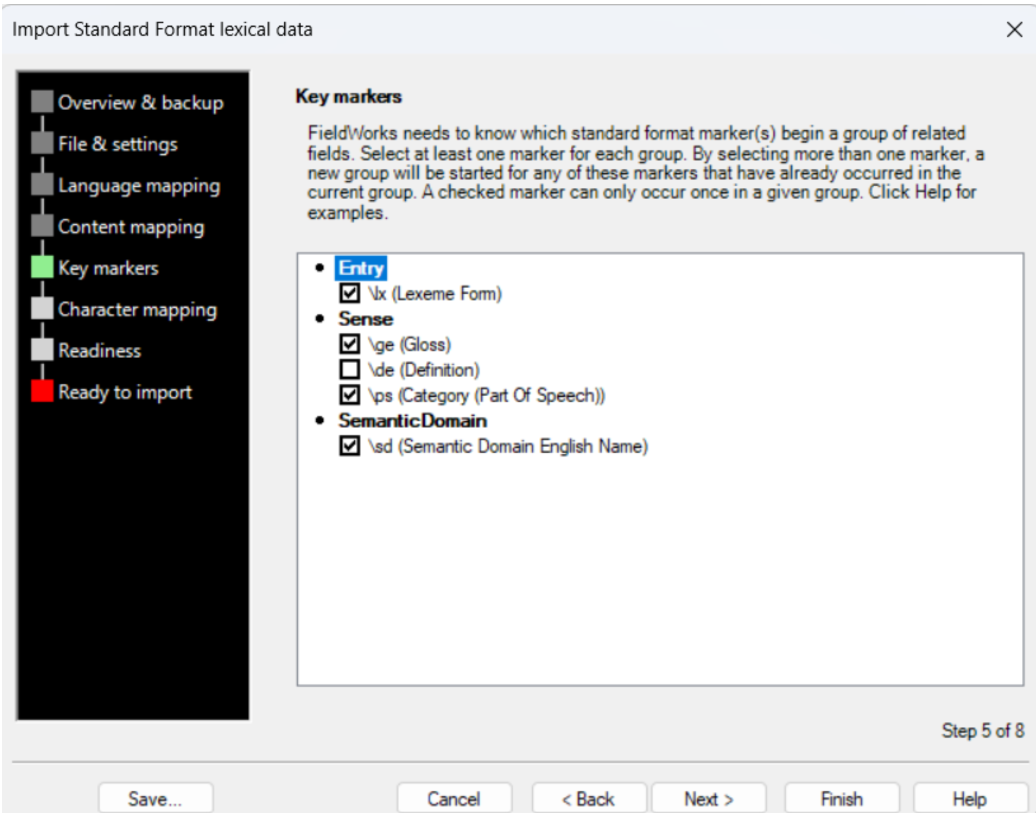
To import the .SFM file, click **File** → **Import**, then select "**Standard Format Lexicon**."

Then:

1. Select your file: *dictionary.sfm* on your computer.
2. Follow the import wizard



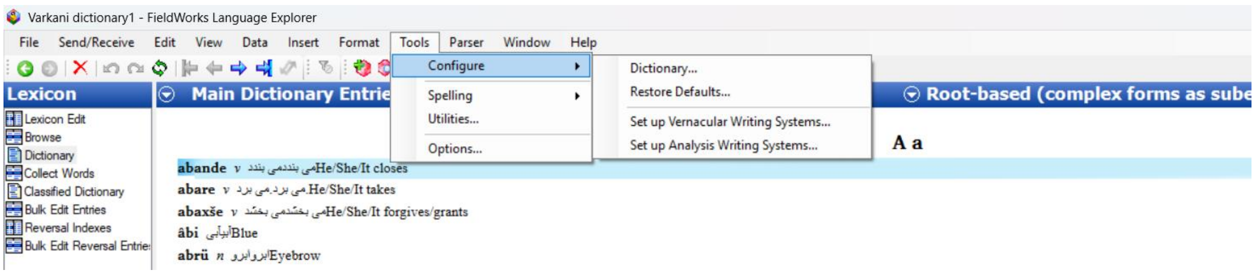
While importing the SFM file, you can modify the key markers and select the options you need.

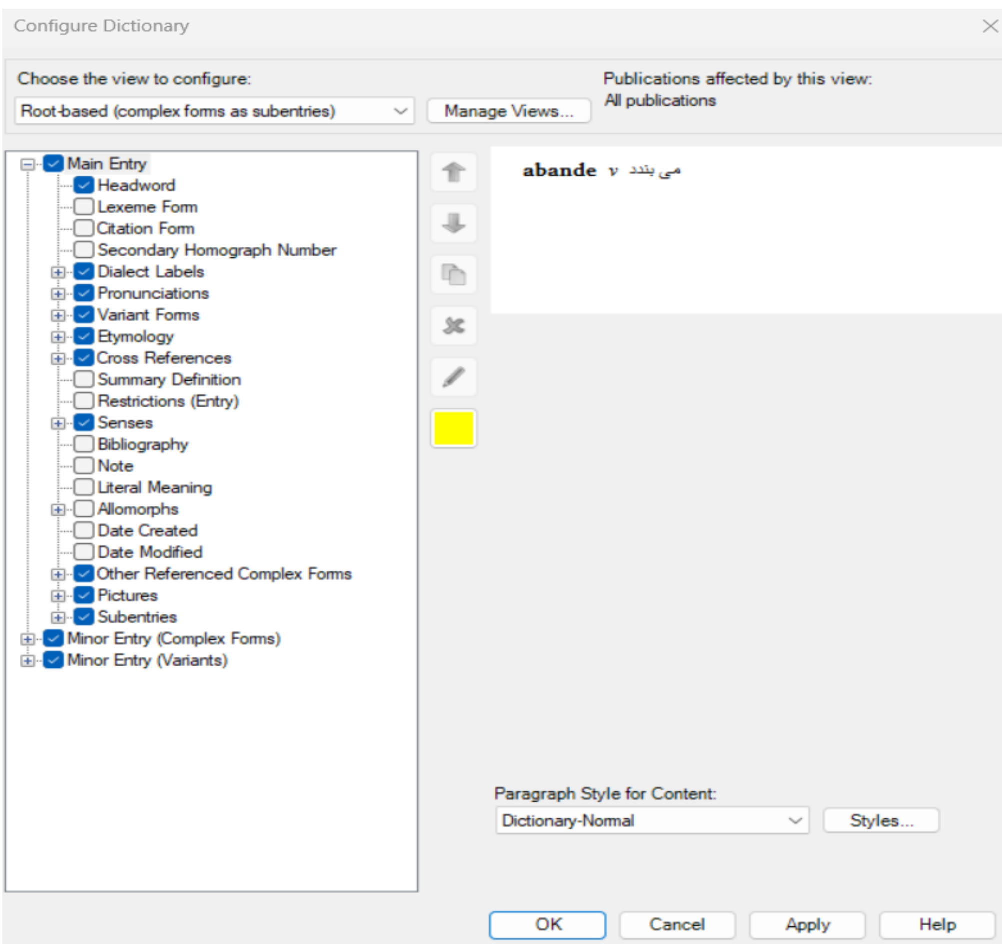


Step 5: Configure languages

Because this is a trilingual dictionary:

- Go to Tools → Configure
- Make sure all three languages are included (Varkani, Persian, English)
- Check that the <u>Sense</u> and <u>Gloss</u> fields are properly assigned





Step 6: Review entries

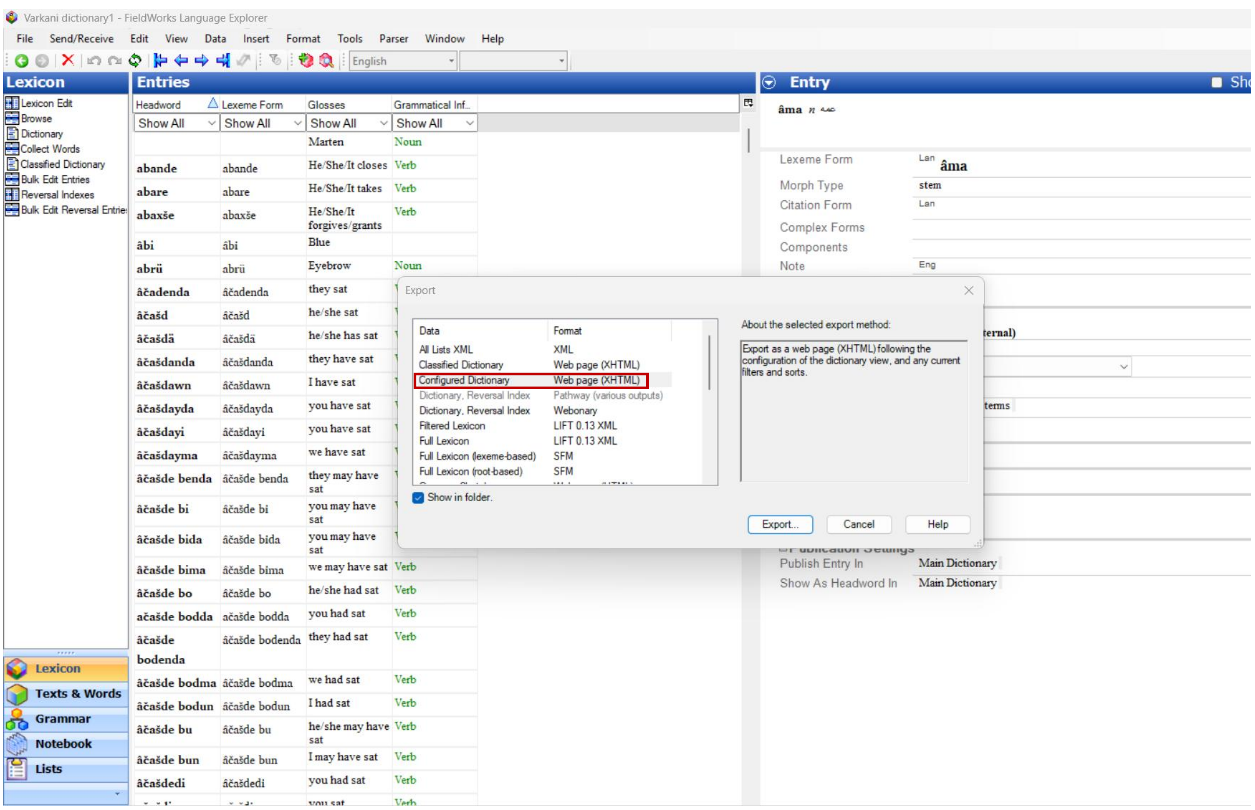
Open the Lexicon view and check:

- Headwords are correct
- Persian translations are correct
- English translations are correct

Fix any issues if needed.

Step 7: Export to HTML

Go to File ⟶ Export ⟶ HTML Dictionary.
Choose an output folder and export the dictionary.



Step 8: Open the dictionary

Open the exported folder and click on index.html.
You now have a searchable HTML dictionary.

A a
abandevمی بنددمی بنددHe/She/It closes
abarevمی برد.می برد.He/She/It takes
abaxševمی بخشدمی بخشدHe/She/It forgives/grants
âbiآبیآبیBlue
abrünnابروابروEyebrow
âčadendavنشستندنشستندthey sat
âčašdvنشستنشستhe/she sat
âčašdävنشسته است.نشسته است.he/she has sat
âčašdandavنشسته اند.نشسته اند.they have sat
âčašdawnvنشسته امنشسته امI have sat
âčašdaydavنشسته ایدنشسته ایدyou have sat
âčašdayivنشسته اینشسته ایyou have sat
âčašdaymavنشسته ایمنشسته ایمwe have sat
âčašde bendavنشسته باشندنشسته باشندthey may have sat
âčašde bivنشسته باشینشسته باشیyou may have sat
âčašde bidavنشسته باشیدنشسته باشیدyou may have sat
âčašde bimavنشسته باشیمنشسته باشیمwe may have sat
âčašde bovنشسته بود.نشسته بود.he/she had sat
ačašde boddavنشسته بودیدنشسته بودیدyou had sat
âčašde bodendavنشسته بودندنشسته بودندthey had sat
âčašde bodmavنشسته بودیمنشسته بودیمwe had sat
âčašde bodunvنشسته بودمنشسته بودمI had sat

I added the HTML file that I exported from FLEx to GitHub.